

T3 UV Lagrangian terms grouped by field content

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. The BSM-only UV expression is preferred; the full UV Lagrangian is used as a fallback when the BSM-only conversion is unavailable.

Field combinations 1 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	$S_1 D$	S_2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-S_1 S_1^\dagger M_{S_1}^2$ $+\frac{1}{2} S_1^2 (S_1^\dagger)^2 \lambda_{S_1}^{(1)}$	$+D_{\mu_1} (S_1)^\dagger D_{\mu_1} (S_1)$	$-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_1,j_3,j_2,j_4}}^1 \lambda_{S_2}^{(1)}$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_1,j_3,j_2,j_4}}^2 \lambda_{S_2}^{(2)}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-S_1 S_1^\dagger M_{S_1}^2$ $+\frac{1}{2} S_1^2 (S_1^\dagger)^2 \lambda_{S_1}^{(1)}$	$+D_{\mu_1} (S_1)^\dagger D_{\mu_1} (S_1)$	$-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_1,j_3,j_2,j_4}}^1 \lambda_{S_2}^{(1)}$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_1,j_3,j_2,j_4}}^2 \lambda_{S_2}^{(2)}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-S_1 S_1^\dagger M_{S_1}^2$ $+\frac{1}{2} S_1^2 (S_1^\dagger)^2 \lambda_{S_1}^{(1)}$	$+D_{\mu_1} (S_1)^\dagger D_{\mu_1} (S_1)$	$-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_1,j_3,j_2,j_4}}^1 \lambda_{S_2}^{(1)}$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_1,j_3,j_2,j_4}}^2 \lambda_{S_2}^{(2)}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-S_1 S_1^\dagger M_{S_1}^2$ $+\frac{1}{2} S_1^2 (S_1^\dagger)^2 \lambda_{S_1}^{(1)}$	$+D_{\mu_1} (S_1)^\dagger D_{\mu_1} (S_1)$	$-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_1,j_3,j_2,j_4}}^1 \lambda_{S_2}^{(1)}$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_1,j_3,j_2,j_4}}^2 \lambda_{S_2}^{(2)}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$-S_1 S_1^\dagger M_{S_1}^2$ $+\frac{1}{2} S_1^2 (S_1^\dagger)^2 \lambda_{S_1}^{(1)}$	$+D_{\mu_1} (S_1)^\dagger D_{\mu_1} (S_1)$	$-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_1,j_3,j_2,j_4}}^1 \lambda_{S_2}^{(1)}$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_1,j_3,j_2,j_4}}^2 \lambda_{S_2}^{(2)}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_1,j_3,j_2,j_4}}^1 \lambda_{S_1}^{(1)}$	$+D_{\mu_1} (S_{1,j_1})^\dagger D_{\mu_1} (S_{1,j_1})$	$-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_1,j_3,j_2,j_4}}^1 \lambda_{S_2}^{(1)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_1,j_3,j_2,j_4}}^1 \lambda_{S_1}^{(1)}$	$+D_{\mu_1} (S_{1,j_1})^\dagger D_{\mu_1} (S_{1,j_1})$	$-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_1,j_3,j_2,j_4}}^1 \lambda_{S_2}^{(1)}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_1,j_3,j_2,j_4}}^1 \lambda_{S_1}^{(1)}$	$+D_{\mu_1} (S_{1,j_1})^\dagger D_{\mu_1} (S_{1,j_1})$	$-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_1,j_3,j_2,j_4}}^1 \lambda_{S_2}^{(1)}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_1,j_3,j_2,j_4}}^1 \lambda_{S_1}^{(1)}$	$+D_{\mu_1} (S_{1,j_1})^\dagger D_{\mu_1} (S_{1,j_1})$	$-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_1,j_3,j_2,j_4}}^1 \lambda_{S_2}^{(1)}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_1,j_3,j_2,j_4}}^1 \lambda_{S_1}^{(1)}$	$+D_{\mu_1} (S_{1,j_1})^\dagger D_{\mu_1} (S_{1,j_1})$	$-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_1,j_3,j_2,j_4}}^1 \lambda_{S_2}^{(1)}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_1,j_3,j_2,j_4}}^1 \lambda_{S_1}^{(1)}$	$+D_{\mu_1} (S_{1,j_1})^\dagger D_{\mu_1} (S_{1,j_1})$	$-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_1,j_3,j_2,j_4}}^1 \lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	D	S_2
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$	$+D_{\mu_1} (S_{1,j_1})^\dagger D_{\mu_1} (S_{1,j_1})$	$-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$
								$+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^1 \lambda_{S_1}^{(1)}$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^2 \lambda_{S_1}^{(2)}$		$+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^1 \lambda_{S_2}^{(1)}$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^2 \lambda_{S_2}^{(2)}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$	$+D_{\mu_1} (S_{1,j_1})^\dagger D_{\mu_1} (S_{1,j_1})$	$-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$
								$+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^1 \lambda_{S_1}^{(1)}$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^2 \lambda_{S_1}^{(2)}$		$+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^1 \lambda_{S_2}^{(1)}$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^2 \lambda_{S_2}^{(2)}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$	$+D_{\mu_1} (S_{1,j_1})^\dagger D_{\mu_1} (S_{1,j_1})$	$-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$
								$+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^1 \lambda_{S_1}^{(1)}$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^2 \lambda_{S_1}^{(2)}$		$+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^1 \lambda_{S_2}^{(1)}$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^2 \lambda_{S_2}^{(2)}$

Field combinations 2 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2 D$	F	$F D$
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+D_{\mu_1} (S_{2,j_1})^\dagger D_{\mu_1} (S_{2,j_1})$	$-\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+D_{\mu_1} (S_{2,j_1})^\dagger D_{\mu_1} (S_{2,j_1})$	$-\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1} (S_{2,j_1})^\dagger D_{\mu_1} (S_{2,j_1})$	$-\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+D_{\mu_1} (S_{2,j_1})^\dagger D_{\mu_1} (S_{2,j_1})$	$-\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+D_{\mu_1} (S_{2,j_1})^\dagger D_{\mu_1} (S_{2,j_1})$	$-\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+D_{\mu_1} (S_{2,j_1})^\dagger D_{\mu_1} (S_{2,j_1})$	$-\overline{F_{r_1}} F_{r_1} M_F$	$+i\overline{F_{r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{r_1})$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+D_{\mu_1} (S_{2,j_1})^\dagger D_{\mu_1} (S_{2,j_1})$	$-\frac{1}{2} F_{r_1}^T C F_{r_1} M_F$	$+\frac{1}{2} i F_{r_1}^T C \gamma^{\mu_1} D_{\mu_1} (F_{r_1})$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+D_{\mu_1} (S_{2,j_1})^\dagger D_{\mu_1} (S_{2,j_1})$	$-\overline{F_{r_1}} F_{r_1} M_F$	$+i\overline{F_{r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{r_1})$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+D_{\mu_1} (S_{2,j_1})^\dagger D_{\mu_1} (S_{2,j_1})$	$-\overline{F_{r_1}} F_{r_1} M_F$	$+i\overline{F_{r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{r_1})$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+D_{\mu_1} (S_{2,j_1})^\dagger D_{\mu_1} (S_{2,j_1})$	$-\overline{F_{r_1}} F_{r_1} M_F$	$+i\overline{F_{r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{r_1})$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+D_{\mu_1} (S_{2,j_1})^\dagger D_{\mu_1} (S_{2,j_1})$	$-\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+D_{\mu_1} (S_{2,j_1})^\dagger D_{\mu_1} (S_{2,j_1})$	$-\frac{1}{2} F_{j_1,r_1}^T C F_{j_1,r_1} M_F$	$+\frac{1}{2} i F_{j_1,r_1}^T C \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+D_{\mu_1} (S_{2,j_1})^\dagger D_{\mu_1} (S_{2,j_1})$	$-\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+D_{\mu_1} (S_{2,j_1})^\dagger D_{\mu_1} (S_{2,j_1})$	$-\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+D_{\mu_1} (S_{2,j_1})^\dagger D_{\mu_1} (S_{2,j_1})$	$-\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+D_{\mu_1} (S_2)^\dagger D_{\mu_1} (S_2)$	$-\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+D_{\mu_1} (S_2)^\dagger D_{\mu_1} (S_2)$	$-\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+D_{\mu_1} (S_2)^\dagger D_{\mu_1} (S_2)$	$-\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+D_{\mu_1} (S_2)^\dagger D_{\mu_1} (S_2)$	$-\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+D_{\mu_1} (S_2)^\dagger D_{\mu_1} (S_2)$	$-\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+D_{\mu_1} (S_{2,j_1})^\dagger D_{\mu_1} (S_{2,j_1})$	$-\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+D_{\mu_1} (S_{2,j_1})^\dagger D_{\mu_1} (S_{2,j_1})$	$-\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1} (S_{2,j_1})^\dagger D_{\mu_1} (S_{2,j_1})$	$-\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+D_{\mu_1} (S_{2,j_1})^\dagger D_{\mu_1} (S_{2,j_1})$	$-\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+D_{\mu_1} (S_{2,j_1})^\dagger D_{\mu_1} (S_{2,j_1})$	$-\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$

Field combinations 3 of 4

[illegible]

[illegible]

[illegible]

Field combinations 4 of 4

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 H$	$S_1 F \ell$	$S_2 F \ell$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+H_{a_1}H_{a_2}S_{1,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger$ $+\overline{\lambda_{T3}}S_2\mathcal{I}_{T3}^*_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$	$+\overline{L_{a_1,i_1}}CPR\overline{F_{j_1,r_1}}^TS_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{y_{1,a_1,j_1},j_1}$ $+\overline{y_{1,i_1,r_1}}L_{a_1,i_1}^TCPLF_{j_1,r_1}\mathcal{I}_{y_{1,a_1,j_1},j_1}S_{1,j_1}^\dagger$ $+\overline{L_{a_1,i_1}}PRF_{j_1,r_1}y_{2,i_1,r_1}\mathcal{I}_{y_{2,a_1,j_1}}S_2^\dagger$	
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+H_{a_1}H_{a_2}S_{1,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger$ $+\overline{\lambda_{T3}}S_2\mathcal{I}_{T3}^*_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$	$+\overline{L_{a_1,i_1}}CPR\overline{F_{j_1,r_1}}^TS_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{y_{1,a_1,j_1},j_1}$ $+\overline{y_{1,i_1,r_1}}L_{a_1,i_1}^TCPLF_{j_1,r_1}\mathcal{I}_{y_{1,a_1,j_1},j_1}S_{1,j_1}^\dagger$ $+\overline{L_{a_1,i_1}}PRF_{j_1,r_1}y_{2,i_1,r_1}\mathcal{I}_{y_{2,a_1,j_1}}S_2^\dagger$	
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+H_{a_1}H_{a_2}S_{1,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger$ $+\overline{\lambda_{T3}}S_2\mathcal{I}_{T3}^*_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$	$+\overline{L_{a_1,i_1}}CPR\overline{F_{j_1,r_1}}^TS_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{y_{1,a_1,j_1},j_1}$ $+\overline{y_{1,i_1,r_1}}L_{a_1,i_1}^TCPLF_{j_1,r_1}\mathcal{I}_{y_{1,a_1,j_1},j_1}S_{1,j_1}^\dagger$ $+\overline{L_{a_1,i_1}}PRF_{j_1,r_1}y_{2,i_1,r_1}\mathcal{I}_{y_{2,a_1,j_1}}S_2^\dagger$	
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+\overline{\lambda_{T3}}S_2,j_2\mathcal{I}_{T3}^*_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ $+H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger$	$+\overline{L_{a_1,i_1}}CPR\overline{F_{j_1,r_1}}^TS_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{y_{1,a_1,j_1},j_1}$ $+\overline{y_{1,i_1,r_1}}L_{a_1,i_1}^TCPLF_{j_1,r_1}\mathcal{I}_{y_{1,a_1,j_1},j_1}S_{1,j_1}^\dagger$ $+\overline{L_{a_1,i_1}}PRF_{j_1,r_1}y_{2,i_1,r_1}\mathcal{I}_{y_{2,a_1,j_1},j_1}S_{2,j_1}^\dagger$	
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\overline{\lambda_{T3}}S_2,j_2\mathcal{I}_{T3}^*_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ $+H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger$	$+\overline{L_{a_1,i_1}}CPR\overline{F_{j_1,r_1}}^TS_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{y_{1,a_1,j_1},j_1}$ $+\overline{y_{1,i_1,r_1}}L_{a_1,i_1}^TCPLF_{j_1,r_1}\mathcal{I}_{y_{1,a_1,j_1},j_1}S_{1,j_1}^\dagger$ $+\overline{L_{a_1,i_1}}PRF_{j_1,r_1}y_{2,i_1,r_1}\mathcal{I}_{y_{2,a_1,j_1},j_1}S_{2,j_1}^\dagger$	
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\overline{\lambda_{T3}}S_2,j_2\mathcal{I}_{T3}^*_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ $+H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger$	$+\overline{L_{a_1,i_1}}CPR\overline{F_{j_1,r_1}}^TS_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{y_{1,a_1,j_1},j_1}$ $+\overline{y_{1,i_1,r_1}}L_{a_1,i_1}^TCPLF_{j_1,r_1}\mathcal{I}_{y_{1,a_1,j_1},j_1}S_{1,j_1}^\dagger$ $+\overline{L_{a_1,i_1}}PRF_{j_1,r_1}y_{2,i_1,r_1}\mathcal{I}_{y_{2,a_1,j_1},j_1}S_{2,j_1}^\dagger$	
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\overline{\lambda_{T3}}S_2,j_2\mathcal{I}_{T3}^*_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ $+H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger$	$+\overline{L_{a_1,i_1}}CPR\overline{F_{j_1,r_1}}^TS_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{y_{1,a_1,j_1},j_1}$ $+\overline{y_{1,i_1,r_1}}L_{a_1,i_1}^TCPLF_{j_1,r_1}\mathcal{I}_{y_{1,a_1,j_1},j_1}S_{1,j_1}^\dagger$ $+\overline{L_{a_1,i_1}}PRF_{j_1,r_1}y_{2,i_1,r_1}\mathcal{I}_{y_{2,a_1,j_1},j_1}S_{2,j_1}^\dagger$	
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\overline{\lambda_{T3}}S_2,j_2\mathcal{I}_{T3}^*_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ $+H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger$	$+\overline{L_{a_1,i_1}}CPR\overline{F_{j_1,r_1}}^TS_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{y_{1,a_1,j_1},j_1}$ $+\overline{y_{1,i_1,r_1}}L_{a_1,i_1}^TCPLF_{j_1,r_1}\mathcal{I}_{y_{1,a_1,j_1},j_1}S_{1,j_1}^\dagger$ $+\overline{L_{a_1,i_1}}PRF_{j_1,r_1}y_{2,i_1,r_1}\mathcal{I}_{y_{2,a_1,j_1},j_1}S_{2,j_1}^\dagger$	